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# Viscoelastic modeling based force control framework for enhancing operation safety of soft tissue for orthopedic surgical robots ☆

Guangming Xia [a], Zifeng Jiang [b], Bin Yao [a], Yu Dai [a,*,1]

[a] Institute of Robotics and Automatic Information Systems, Nankai University, Tianjin 300350, China
[b] Viterbi School of Engineering, University of Southern California, Los Angeles, CA 90007, United States

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ABSTRACT

In robot-assisted laminectomy, precise control of touch force on the spinal cord during the dissection of ossified ligaments is critical to prevent dura mater tearing and ensure surgical safety. However, the complex viscoelastic characteristics of tissue at the lesion site pose a significant challenge for accurately applying touch force. This paper proposes a safe touch force servo framework based on tissue model identification and preoperative optimization of the force controller. First, a soft tissue model is established using elastic and viscoelastic elements, and the transfer function from tissue deformation to touch force is derived. Subsequently, the path and parameters of a safe pre-touch are designed, and force information required for tissue model identification is obtained through the pre-touch experiment. The model order and specific parameter values are then identified using a differential evolution algorithm. Next, force control simulation is conducted based on the identified model, and a specifically designed loss function is introduced to achieve preoperative tuning of the force controller. Finally, step force control experiments on sheep spines validate the effectiveness of the proposed method. This paper provides a safe and available touch motion control framework that can be expandable to impose precise force on various vulnerable soft tissues, with the potential for widespread application.

## 1. Introduction

### 1.1. Research background and motivation

The function of the spine is vital to the body health and daily activities [1]. However, changes in modern work lifestyles have led to an increase in various spine disorders, particularly those related to nerve compression [2]. Laminectomy is a common surgical procedure aimed at relieving nerve compression in spinal disorders, involves removing the lamina to expose the ossified posterior longitudinal ligament [3,4]. Subsequently, the ossified portion is dissected to release spinal canal space, alleviating nerve compression in patients [5]. In traditional surgeries, surgeons manually manipulate tools to dissect the ossified ligament, and preventing dura mater tears is crucial for ensuring surgical safety [6]. However, traditional surgeries are constrained by factors such as physiological hand tremors, fatigue from prolonged procedures, and lack of proficiency, all of which may result in damage to vital neural tissues [7]. Utilizing robot assistance in laminectomy and dissection procedures is considered a solution to these challenges, enhancing overall surgical safety [8,9]. In robot-assisted procedures for dissecting ossified ligaments, precise control of the interaction force between the robot and the spinal cord is crucial to prevent excessive dissection force from tearing the dura mater. This necessitates improved automatic control of the interaction force between the robot and the spinal cord.

### 1.2. Related works

Soft tissue touch force information has been widely applied in studies such as tumor palpation [10,11]. The class of the soft tissue can be identified because of their different mechanical properties [12,13]. Considering complex viscoelastic mechanical properties of some tissues, some researchers selected to use finite element analysis (FEA) for detailed modeling [14–16]. However, the FEA method involves significant computation and time duration, and the established FEA model cannot be further used for the iterative optimization of force controller parameters. Collective parametric models are commonly used

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* Corresponding author.
*E-mail addresses:* xiaguangming@mail.nankai.edu.cn (G. Xia), zifengji@usc.edu (Z. Jiang), daiyu@nankai.edu.cn (Y. Dai).
1 Member, IEEE.

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for modeling tissues due to their simplicity, fast response time, and relatively realistic deformations [17]. For instance, the three-parameter principal model is widely used to roughly represent the viscoelastic behavior of different types of soft tissues [18]. Considering that the three-parameter model cannot accurately characterize the mechanical properties of some complex viscoelastic tissues, a parallel model with elastic and Maxwell units was introduced in this study to characterize the tissue composed of the dura mater and spinal cord. This ensures both the accuracy of the model and its suitability for closed-loop simulation, thereby making the established model applicable for the iterative optimization of the force controller.

There are also some related researches focusing on the development of perception technology. For example, some improved force sensors [12,19] and various vision-based force estimation methods [20,21] are designed to better perceive touch force and improve the efficiency and recognition accuracy of palpation. For example, Y. E. Lee et al. utilized deep learning, relying solely on visual information and inertial data from the instrument to estimate palpation force [20]. W.C. Yue et al. proposed presents a novel dynamic tactile sensor with a symmetrical longitudinal piezoelectric cantilever structure [12]. The commonality among these studies lies in the identification of soft tissue categories based on force data and instrument's location information, with some specific categories potentially involving the location and developmental stage of tumor tissues. The difference lies in the specific processing and computational procedures related to force information acquisition due to task variations. During the laminectomy procedure, the touch force on the dura mater can be directly measured by force sensors installed on the mechanical unit executing the pressing action. Therefore, this study utilizes directly measured force for feedback rather than relying on estimated force based on vision, electricity or other indirect methods.

Most soft tissue touch force control studies focus on designing controllers and verifying stability, aiming to ensure that surgical tools can achieve smooth and precise force transmission when in contact with soft tissues [22–25]. For instance, J. Guo employed a fuzzy PID controller in a vascular intervention surgical robot [22]. Q.Y. Ren introduced a neural network-based surgical robot force controller, incorporating a compensator based on adaptive proportional–integral-sliding-mode control [25]. These studies all emphasize that well-designed instrument force control can enhance surgical safety and potentially alleviate postoperative pain for patients. However, tuning the parameters of force controllers based on experience is challenging. Therefore, this paper aims to validate a preoperative controller tuning framework, using a small amount of safely acquired touch force information to adjust the controller parameters. This study demonstrates the proposed method in the context of controlling touch force on the dura mater of sheep spines, and the presented framework is scalable. The optimization algorithm and controller within the framework can be replaced according to the requirements of other relevant tasks.

### 1.3. Contributions and organization of this study

The contributions are mainly in the following aspects:

(1) Proposed a force servo framework for robotic safe touch based on tissue model identification and preoperative optimization of force controllers;

(2) Developed the soft tissue model based on elastic and viscoelastic elements and derived its transfer function from tissue deformation to touch force;

(3) Designed the path and parameters for a safe preliminary touch, and obtained the force information required for tissue model identification through experiments;

(4) Model identification and force control simulations were performed, and a specially designed loss was introduced to enable preoperative adjustment of the force controller.

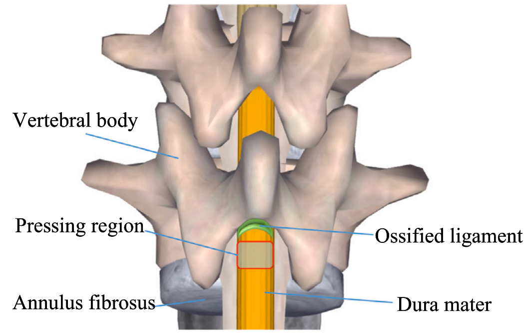



**Fig. 1.** Schematic diagram of SCS caused by ligament ossification.

(5) The validity of the proposed method was verified by performing force control experiments on a sheep spine.

The rest of this article is organized as follows. In Section 2, the proposed method is introduced in detail. In Section 3, the experiments, results and discussion are provided. The conclusion and future plans are provided in Section 4.

## 2. Methods

### 2.1. Problem description

Laminectomy is the most common orthopedic surgery in clinics used to treat spinal canal stenosis (SCS) by relieving compression on the nerves in the spine. Fig. 1 illustrates the schematic diagram of the spine when SCS is caused by the ossification of the posterior longitudinal ligament. As depicted in Fig. 1, the ossified posterior longitudinal ligament (highlighted in green) compresses the spinal nerve inwardly. To alleviate nerve compression, orthopedists need to cut off the spinous processes and lamina of the vertebral body, subsequently exposing and dissecting the posterior longitudinal ligament. During the process of cutting off the spinous processes and lamina and dissecting the ossified ligament, an assistant may often need to gently press the dura mater of the patient's spinal cord to facilitate the dissection of the ossified ligament and prevent tearing of the dura mater. The relative position of the press contraction area of the spinal cord and the ossified ligament is indicated in Fig. 1. In robot-assisted surgery, robotic arm is hoped to apply pressure automatically. However, the viscoelastic properties of the soft tissue around the lesion area can vary greatly between patients, posing a challenge in determining universal controller parameters. Therefore, it is crucial to develop a control framework capable of accurately identifying the soft tissue model and automatically optimizing the force controller parameters. This will enable the robot to consistently achieve the best possible press force servo, ensuring safe and effective surgery for different patients.

### 2.2. Soft tissue's force model and transfer function establishment

This section aims to develop a transfer function to characterizes the relationship between the applied deformation on the soft tissue and the resulting contact force. Fig. 2(a) describes the schematic diagram of the contacting process between the surgical tool and the spinal cord. The arc-shaped end of the tool is in direct contact with the spinal cord, and the blue dotted line represents the deformed spinal cord after pressing. Given that the dura mater exhibits linear elasticity while the spinal cord displays complex viscoelastic properties, Fig. 2(b) depicts a soft tissue model that comprises a single elastic element and several Maxwell elements arranged in parallel. The elastic element is utilized to capture the behavior of linear elasticity, while the complex viscoelasticity is represented using several Maxwell elements. The Maxwell element is composed of a damping element and an elastic element in series. Let

$x(t)$ denotes the displacement of arc-shaped end after contacting the spinal cord ($x(t)$ is also the applied deformation on the soft tissue), and $f(t)$ denotes the resulting contact force. Perform force analysis on the soft tissue model

$$f(t) = f_0(t) + \sum_{i=1}^{n} f_i(t) \tag{1}$$

where $f_0(t)$ denotes the linear elastic force, $f_i(t)$ denotes the viscoelastic force from the $i$th Maxwell element.

Through force analysis, it can be obtained that the geometric relationship between the elastic and damping elements inside the $i$th Maxwell element is

$$x(t) = x_i^{\mathrm{e}}(t) + x_i^{\mathrm{d}}(t) \tag{2}$$

where $x_i^{\mathrm{e}}(t)$ denotes the deformation amount of the elastic element inside the Maxwell element, and $x_i^{\mathrm{d}}(t)$ denotes the deformation amount of the damping element inside the Maxwell element. Derived from (2), further get

$$\dot{x}(t) = \dot{x}_i^{\mathrm{e}}(t) + \dot{x}_i^{\mathrm{d}}(t) \tag{3}$$

Further, analyzing the internal force of the $i$th Maxwell element, as follows

$$f_i(t) = f_i^{\mathrm{e}}(t) = f_i^{\mathrm{d}}(t) \tag{4}$$

$$f_i^{\mathrm{e}}(t) = k_i x_i^{\mathrm{e}}(t) \tag{5}$$

$$f_i^{\mathrm{d}}(t) = c_i \dot{x}_i^{\mathrm{d}}(t) \tag{6}$$

where $k_i$ and $f_i^{\mathrm{e}}(t)$ represent the stiffness coefficient and output force of the elastic element inside the Maxwell element, respectively. And $c_i$ and $f_i^{\mathrm{d}}(t)$ represent the damping coefficient and output force of the internal damping element of the Maxwell element, respectively. According to (4), (5) and (6), (7) and (8) can be expressed as follows

$$\dot{x}_i^{\mathrm{e}}(t) = \frac{\dot{f}_i(t)}{k_i} \tag{7}$$

$$\dot{x}_i^{\mathrm{d}}(t) = \frac{f_i(t)}{c_i} \tag{8}$$

According to (3), (7) and (8), (9) can be obtained

$$f_i(t) + \frac{c_i}{k_i}\dot{f}_i(t) = c_i \dot{x}(t) \tag{9}$$

and the force of the elastic element $k_0$ can be calculated as

$$f_0(t) = k_0 x(t) \tag{10}$$

Carry out Laplace transform on (9) and (10), and substitute them into (1), (11) is further obtained as follows

$$f(s) = (k_0 + \sum_{i=1}^{n} \frac{c_i s}{(c_i/k_i)s + 1})x(s) \tag{11}$$

**Conclusion**: a transfer function, which can characterize the relationship between the applied deformation on the soft tissue and the resulting contact force, is established in this section.

### 2.3. Safe pre-touch design and model identification

This section aims to identify the parameters associated with the soft tissue model established in Section 2.2 effectively based on the force response data obtained from a specially designed touch experiment named safe pre-touch. During pre-touch procedure, a specific strain is carried out to the soft tissue. Simultaneously, recording how stress changes over time. To apply a safe strain, the acceleration of the surgical instrument is planned as follows

$$a(t) = \begin{cases} A\sin(f_a t) & t < \frac{1}{f_a} \\ 0 & t > \frac{1}{f_a} \end{cases} \tag{12}$$

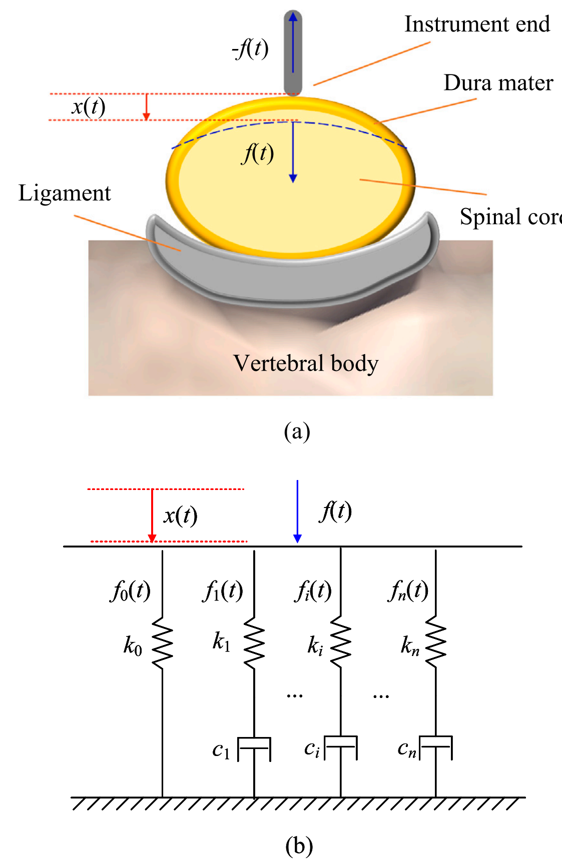



**Fig. 2.** Soft tissue model establishment. (a) the contact between the instrument and the dura mater; (b) Soft tissue contact force model.

where $A$ and $f_a$ are the amplitude and frequency of the sinusoidal acceleration. The starting speed is zero, and the starting position of the instrument's end is the surface of the dura mater. Therefore, the displacement of the instrument is equal to the deformation of the target soft tissue, as follows

$$x(t) = \frac{-A}{(2\pi f_a)^2}\sin(2\pi f_a t) + \frac{A}{2\pi f_a}t \tag{13}$$

Based on (13), when time is greater than $1/f_a$, the tissue deformation reaches its maximum value of $A/2\pi$. The maximum deformation can be controlled by adjusting $A$ and the time required for the tissue deformation to reach its maximum can be controlled by adjusting $f_a$.

The common driving method for surgical instruments is based on discrete digital speed servo. Therefore, the input of the surgical instrument at time $k$ is a discrete digital speed

$$v(k) = [x(k+1) - x(k)]f_{\mathrm{co}} \tag{14}$$

where $f_{\mathrm{co}}$ represents the transmission frequency of speed command for the surgical instrument's driving motor. $x(k)$ and $x(k+1)$ are the planned ideal positions of the instrument's end at time $k$ and $k+1$, respectively.

In a safe pre-touch test, a force sensor are used to collect corresponding force response. Assuming the force sensor's sampling frequency is $f_{\mathrm{F}}$, and the collection duration is $T_{\mathrm{s}}$, the obtained force signal is

$$\mathbf{F}_{\mathrm{e}} = [F_{\mathrm{e}}(1) \quad F_{\mathrm{e}}(2) \quad \cdots \quad F_{\mathrm{e}}(T_{\mathrm{s}}/f_{\mathrm{F}})] \tag{15}$$

Assuming a simulation frequency of $f_{\mathrm{sim}}$, and a duration equal to $T_{\mathrm{s}}$, for each set of solution vectors $\Delta = [k_0, I, k_1, \ldots, k_I, c_1, \ldots, c_I]$ inserted into the corresponding transfer function of (11) and provided with the same input velocity, a set of simulated touch forces can be obtained.

$$\mathbf{F}_{\mathrm{sim}} = [F_{\mathrm{sim}}(1) \quad F_{\mathrm{sim}}(2) \quad \cdots \quad F_{\mathrm{sim}}(T_{\mathrm{s}}/f_{\mathrm{sim}})] \tag{16}$$

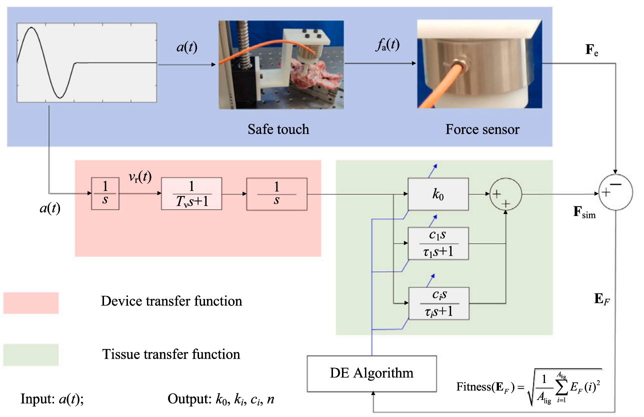



**Fig. 3.** Flow chart of soft tissue model parameter identification based on stress relaxation tests and DE algorithm.

The key to identifying system parameters based on optimization algorithms lies in the calculation of the fitness function. Since the sampling frequencies of the two forces may differ, interpolation is used to align the lengths of $F_{\mathrm{sim}}$ and $F_{\mathrm{e}}$ before computing the error. Assuming the aligned length is $A_{\mathrm{lig}}$, a fitting error vector $E_{\mathrm{F}}$ is then obtained

$$\begin{aligned}\mathbf{E}_F &= [(F_{\mathrm{sim}}(1) - F_{\mathrm{e}}(2)) \quad \cdots \quad (F_{\mathrm{sim}}(A_{\mathrm{lig}}) - F_{\mathrm{e}}(A_{\mathrm{lig}}))] \\ &= [E_F(1) \quad E_F(1) \quad \cdots \quad E_F(A_{\mathrm{lig}})]\end{aligned} \tag{17}$$

And the calculation of the fitness function during the identification of system parameters is

$$\mathrm{Fitness}(\mathbf{E}_F) = \sqrt{\frac{1}{A_{\mathrm{lig}}}\sum_{i=1}^{A_{\mathrm{lig}}} E_F(i)^2} \tag{18}$$

Given that differential evolution algorithms address complex optimization problems through cooperation and competition among individuals within a population, they exhibit strong robustness and global optimization capabilities. Therefore, this study employs the differential evolution (DE) algorithm for subsequent system parameter identification. Fig. 3 summarizes the parameter identification process.

It is important to note that optimization algorithms are not unique. As long as the model identification error meets the requirements of practical applications, alternative algorithms may also be considered, such as particle swarm optimization (PSO) in [26] or whale optimization algorithm (WOA) in [27].

### 2.4. Touch force controller and optimization procedure

Optimizing the controller parameters lies in the calculation of the objective function. In order to obtain satisfactory dynamic performance of the transition process, the integral time absolute error is used as the basic objective function for parameter selection. To mitigate the risk of the surgical instrument's feed speed being excessively high, a penalty term for the feed speed (control input) is incorporated into the objective function. Furthermore, an overshoot penalty term is introduced to minimize overshooting. In summary, the formula for calculating the objective function J is as follows:

$$\begin{aligned}&\text{if}: \quad (u(t) < u_{\mathrm{m}}) \vee (e(t) > 0) \\ &J = \int_0^{T_s} (w_1\,|e(t)|)tdt \\ &\text{else}: \\ &J = \int_0^{T_s} (w_1\,|e(t)| + w_2\,|e(t)|)tdt\end{aligned} \tag{19}$$

where $T_{\mathrm{s}}$ is the simulation time, $w_1$, and $w_2$ are the weight coefficients of error term and punishment term respectively. $e(t)$ is the deviation between the actual contact force and the expected force. $u(t)$ is the feed speed of the end of the surgical instrument, and um is the allowed max input of $u(t)$.

To prove the generality of the proposed method, proportion integration differentiation (PID), linear active disturbance rejection control (LADRC), and internal model control (IMC) controllers are employed for performance testing. These three controllers represent model-free, model-estimated, and model-based approaches. Fig. 4 shows their control frameworks. Next, they are briefly introduced, and their parameters requiring tuning in practice are determined.

The basic PID controller transfer function is

$$C_{\mathrm{PID}}(s) = k_{\mathrm{p}} + \frac{k_{\mathrm{i}}}{s} + k_{\mathrm{d}}s \tag{20}$$

where $k_{\mathrm{p}}$, $k_{\mathrm{i}}$ and $k_{\mathrm{d}}$ are the proportional, integral and the differential coefficients respectively.

The basic LADRC controller has a linear extended state observer (LESO). The LESO applied in soft tissue contact force control is as follows

$$\begin{aligned}\dot{x} &= Ax + Bu, \\ y &= Cx + Du,\end{aligned} \tag{21}$$

where

$$A = \begin{bmatrix} -3\omega_{\mathrm{o}} & 1 & 0 \\ -3{\omega_{\mathrm{o}}}^2 & 0 & 1 \\ -{\omega_{\mathrm{o}}}^3 & 0 & 0 \end{bmatrix}, B = \begin{bmatrix} 0 & 3\omega_{\mathrm{o}} \\ 1 & 3{\omega_{\mathrm{o}}}^2 \\ 0 & {\omega_{\mathrm{o}}}^3 \end{bmatrix} \tag{22}$$

$$C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, D = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \tag{23}$$

where $w_{\mathrm{o}}$ is defined as the observer bandwidth frequency, and input and output vectors of the above LESO are

$$u = [v \quad F_{\mathrm{c}}]^{\mathrm{T}} \tag{24}$$

$$y = [z_1 \quad z_2 \quad z_3]^{\mathrm{T}} = [F_{\mathrm{c}} \quad \dot{F}_{\mathrm{c}} \quad \ddot{F}_{\mathrm{c}}]^{\mathrm{T}} \tag{25}$$

Moreover, the controller bandwidth frequency $\omega_{\mathrm{c}}$ is defined as follows

$$\omega_{\mathrm{c}} = o_1 \times \omega_{\mathrm{o}} \tag{26}$$

where suggested value range of $o1$ is 0.5 to 1. The proportional and differential coefficients $k_{\mathrm{p}}$ and $k_{\mathrm{d}}$ in $C_{\mathrm{PID}}(s)$ are calculated as follows

$$k_{\mathrm{p}} = o_{\mathrm{p}} \times {w_{\mathrm{c}}}^2 \tag{27}$$

$$k_{\mathrm{d}} = 2 \times o_{\mathrm{d}} \times w_{\mathrm{c}} \tag{28}$$

The coefficients $o_{\mathrm{p}}$ and $o_{\mathrm{d}}$ is used to fine-tune $k_{\mathrm{p}}$ and $k_{\mathrm{d}}$, and their ranges should be set by experience.

In IMC design, the transfer function of the controlled object is needed as reference. Assume the contact force model of the target soft tissue has been identified as follows:

$$G_{\mathrm{e}}(s) = \frac{f(s)}{x(s)} = \hat{k}_0 + \frac{\hat{c}_1 s}{\hat{\tau}_1 s + 1} + + \frac{\hat{c}_2 s}{\hat{\tau}_2 s + 1} \tag{29}$$

and $G_{\mathrm{e}}(s)$ can be obtained

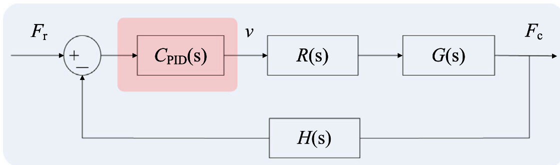
$$G_{\mathrm{e}}(s) = \frac{q_2 s^2 + q_1 s + q_0}{\hat{\tau}_1\hat{\tau}_2 s^2 + (\hat{\tau}_1 + \hat{\tau}_2)s + 1} \tag{30}$$

where

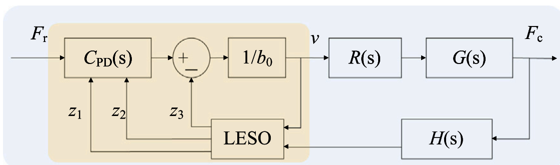
$$q_2 = \hat{k}_0\hat{\tau}_1\hat{\tau}_2 + \hat{c}_1\hat{\tau}_2 + \hat{c}_2\hat{\tau}_1 \tag{31}$$

$$q_1 = \hat{k}_0(\hat{\tau}_1 + \hat{\tau}_2) + \hat{c}_1 + \hat{c}_2 \tag{32}$$

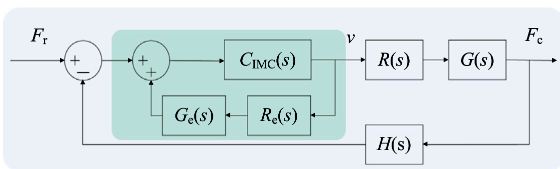
$$q_0 = \hat{k}_0 + \hat{c}_1 + \hat{c}_2 \tag{33}$$




(a) PID controller based force control framework




(b) LADRC controller based force control framework




(c) IMC controller based force control framework

**Fig. 4.** Three control frameworks: model-free, model-estimated, and model-based approaches. (a) PID controller-based force control framework; (b) LADRC controller-based force control framework; (c) IMC controller-based force control framework. The shaded area highlights the controller part.

Assume that the identified transfer function $R_e(s)$ of instrument drive device is as follows

$$R_e(s) = \frac{x(s)}{v(s)} = \frac{1}{s(\hat{t}_m s + 1)} \tag{34}$$

Then there is a reverse function $G^{-}(s)$

$$G^{-}(s) = \frac{1}{G_e(s)R_e(s)} = \frac{(t_m s^2 + s)(\hat{\tau}_1 \hat{\tau}_2 s^2 + (\hat{\tau}_1 + \hat{\tau}_2)s + 1)}{q_2 s^2 + q_1 s + q_0} \tag{35}$$

And the IMC controller $C_{IMC}(s)$ can be obtained

$$C_{IMC}(s) = G^{-}(s)F(s) \tag{36}$$

where

$$F(s) = \frac{1}{(a_1 s + 1)(a_2 s + 1)\dots(a_u s + 1)} \tag{37}$$

Considering physical achievability and sensor delay of this study, $F(s)$'s order is at least second order.

**Conclusion**: The parameters of PID that should be optimized are $k_p$, $k_i$ and $k_d$. The parameters of LADRC that should be optimized are $w_o$, $o_1$, $o_p$ and $o_d$, and then the optimal $w_c$, $k_p$ and $k_d$ can be further calculated via (26), (27), and (28). The parameter of IMC that should be optimized is $a_1, a_2, \dots, a_u$.

## 3. Experiments, results and discussions

### 3.1. Experimental hardware and specimen preparation

The hardware for soft tissue's contact force model identification experiment is shown in Fig. 5: A plastic rod with rounded end is utilized for applying pressure on the surface of the dura mater. The motion of the plastic rod is actuated by a rotary motor and a linear guide. A force sensor is positioned between the guide platform and the plastic rod. The demo object is the spinal cord of the sheep spine. A metal vise is utilized for fixing the spine specimen. The spinous process and lamina are removed to fully expose the spinal cord.

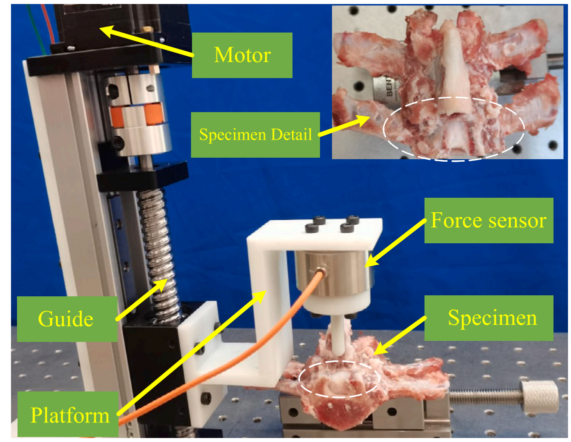



**Fig. 5.** Experimental hardware and spine specimen.

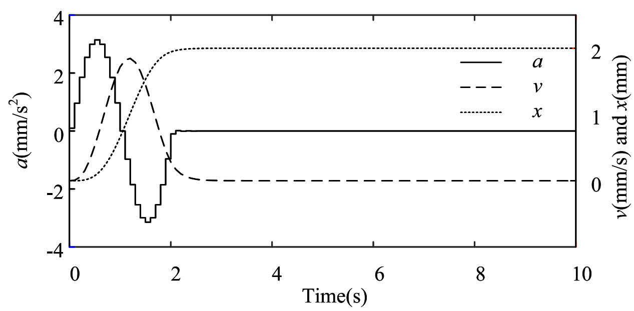



**Fig. 6.** Surgical instrument movement curves.

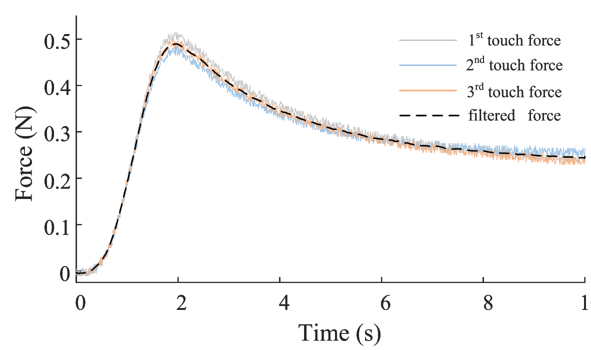



**Fig. 7.** Three sets of force captured by the sensor and the smoothed force.

### 3.2. Force data obtaining and model identification

Use the method in Section 2.3 for obtaining force data. The sampling frequency $f_s$ of the force sensor is set to 100 Hz. The maximum displacement is limited in 3 mm to ensure safety by setting amplitude $A$ of the acceleration as 3.14 mm/s$^2$. The acceleration, velocity, and displacement curves of the surgical instrument movement over time are shown in Fig. 6, and Fig. 7 shows the force curve captured by the force sensor. Further, this section employed a system identification based on the methods in Section 2.3 and the force response data. Detailed algorithm parameter settings during the system identification process and the identification results will be provided. The number $n$ of viscoelastic units determines the order of the system. Hence, identification should initially be conducted with a low order, and if the accuracy does not meet the requirements, the order is increased.

Table 1 provides the DE algorithm parameter settings for $n = 1$, including the population size $N_p$, scaling factor $F$, crossover probability

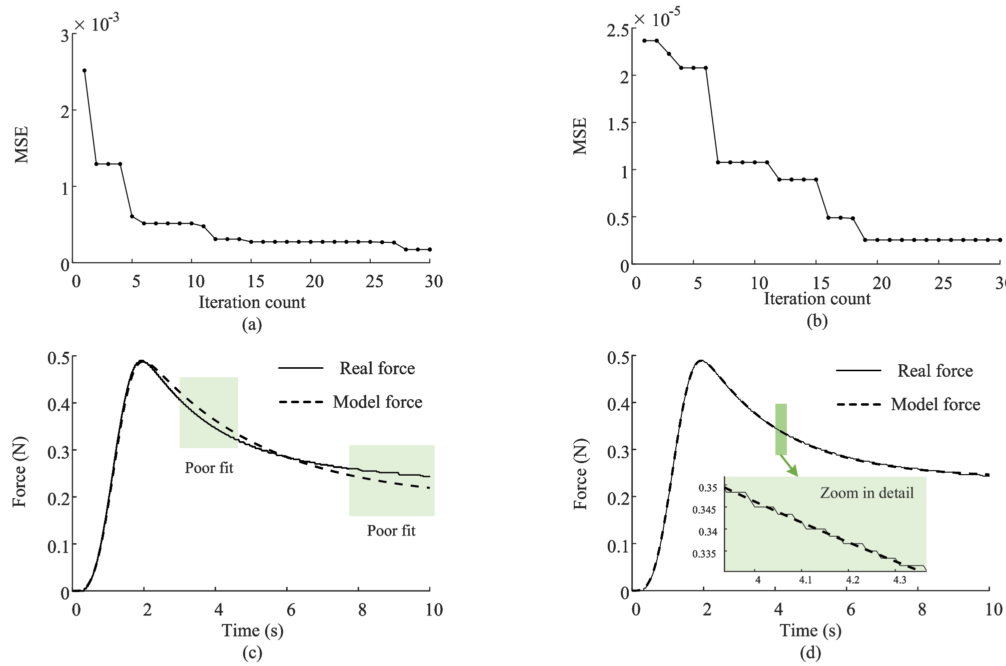



**Fig. 8.** Model identification results. (a) Convergence curve of loss when $n = 1$; (b) Convergence curve of loss when $n = 2$; (c) Comparison of model force and actual force when $n = 1$; (d) Comparison of model force and actual force when $n = 2$.

**Table 1**
Parameter settings and results for system identification when using DE algorithm and supposing $n = 1$.

| Parameter | Value | Parameter | Value | Range |
|---|---|---|---|---|
| $N_p$ | 12 | $k_0$ | 0.096 | [0.04, 0.12] |
| $F$ | 0.5 | $c_1$ | 0.654 | [0.20, 2.00] |
| $C_r$ | 0.3 | $\tau_1$ | 3.298 | [1.00, 18.0] |

**Table 2**
Parameter settings and results for system identification when using DE algorithm and supposing $n = 2$.

| Parameter | Value | Parameter | Value | Range |
|---|---|---|---|---|
| $N_p$ | 20 | $k_0$ | 0.061 | [0.04, 0.08] |
| $F$ | 0.5 | $c_1$ | 0.394 | [0.20, 0.40] |
| $C_r$ | 0.3 | $\tau_1$ | 13.12 | [10.0, 18.0] |
| – | – | $c_2$ | 2.122 | [1.50, 2.50] |
| – | – | $\tau_2$ | 209.5 | [190 , 210] |

$C_r$, final parameter's value and their corresponding optimization range. The convergence curve of the loss function for $n = 1$ is shown in Fig. 8(a). Fig. 8(c) compares the force response of the model with real force data for $n = 1$. It can be observed that the fitted model does not exhibit satisfactory accuracy in the force decay phase, prompting an increase in $n$ for further identification.

Table 2 provides the DE algorithm parameter settings for $n = 1$, including the population size $N_p$, scaling factor $F$, crossover probability $C_r$, final parameter's value and their corresponding optimization range. The convergence curve of the loss function is depicted in Fig. 8(b). Fig. 8(d) compares the force response of the model with real force data for $n = 2$, revealing that the current error is only due to noise, and there is no trend error. The fitted model has a sufficient order and good accuracy, making it suitable for tuning the parameters of the force controller in subsequent simulations. Increasing the order of viscoelastic elements can improve the model's fitting accuracy. However, a higher order also increases simulation time, making it unsuitable for real-time use. Based on our experiments, second-order viscoelastic elements offer a good balance of accuracy and efficiency, meeting the needs of real-time applications.

**Table 3**
Parameter settings for PID controller optimization and the optimized parameters' value.

| Parameter | $k_p$ | $k_i$ | $k_d$ |
|---|---|---|---|
| Optimized value | 15.26 | 0.53 | 0.99 |
| Range setting | [0, 50] | [0, 1] | [0, 5] |

**Table 4**
Parameter settings for LADRC controller optimization and the optimized parameters' value.

| Parameter | $w_o$ | $o_1$ | $o_p$ | $o_d$ |
|---|---|---|---|---|
| Optimized value | 9.98 | 0.51 | 4.96 | 0.72 |
| Range setting | [1, 12] | [0.5, 1] | [1, 20] | [0, 4] |

### *3.3. Controller parameter tuning, analysis and practical testing*

This section provides a detailed overview of the process and results of automatically optimizing PID controllers, LADRC controllers, and IMC controllers using the model presented in Table 2. The adopted loss is calculated according to (19), and the parameters are kept consistent ($w_1 = 1$, $w_2 = 100$, $u_m = 1.5$ mm/s, and $T_s = 10$ s) when tuning these three controllers. Tables 3, 4, and 5 present the tuning results for the three controllers. Additionally, Fig. 9(a) illustrates the convergence curves of the loss function for these three controllers during the tuning process. From Fig. 9(a), it is evident that under the same loss function setting, the IMC controller converges faster compared to PID and LADRC controllers. In other words, IMC approaches convergence to the optimal state more rapidly, especially in the early stages of adjustment. It is noteworthy that after a sufficient number of iterations, the difference in the loss values between the IMC controller and the well tuned PID or LADRC controllers are not significant. This implies that when the model of the controlled object is known, PID and LADRC controllers can achieve good control performance through parameter tuning. This emphasizes that accurately establishing and identifying the deformation-force model of soft tissues is crucial for achieving better controller parameter tuning.

Furthermore, compared to IMC, PID, and LADRC require more iterations ($I_a$) for adjustment (In our test, under the same loss, IMC

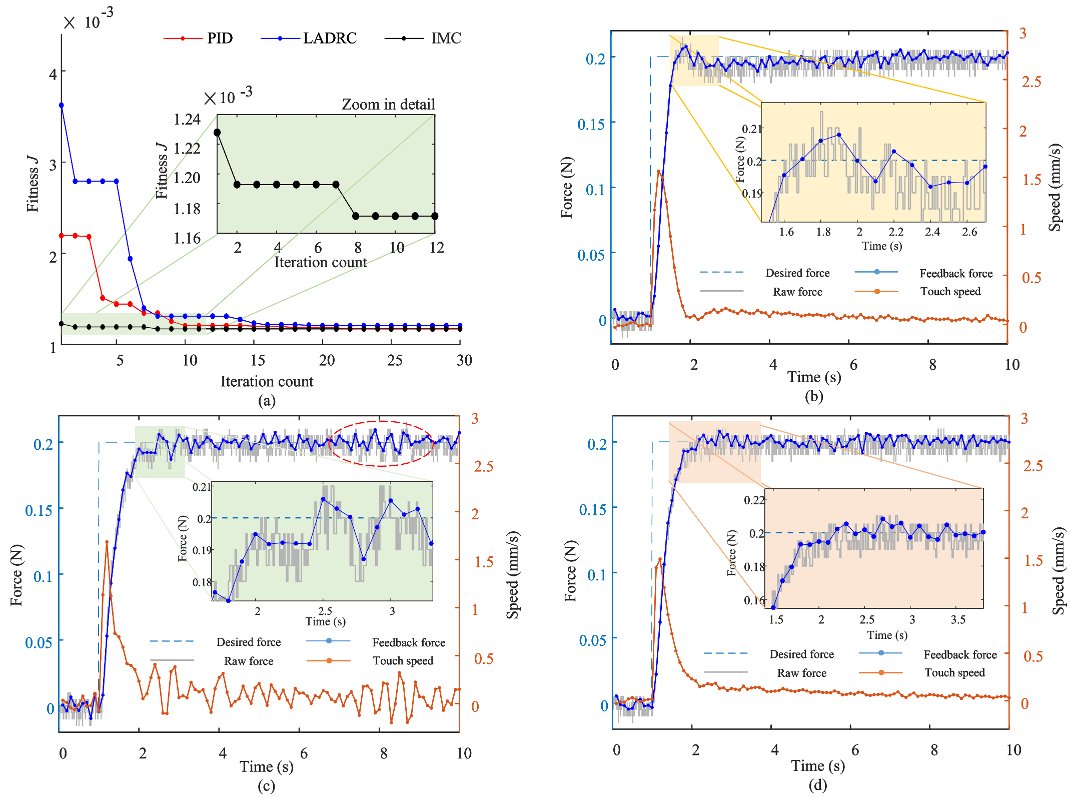


Fig. 9. Experiment results on spinal cord of sheep spine specimen. (a) Convergence curves of loss function $J$ during controller parameter tuning period via simulation. (b) Typical touch force control result with optimized PID. (c) Typical touch force control result with optimized LADRC. (d) Typical touch force control result with optimized IMC.

**Table 5**
Parameter settings for IMC controller optimization and the optimized parameters' value.

| Parameter | $a_1$ | $a_2$ |
|---|---|---|
| Optimized value | 0.255 | 0.084 |
| Range setting | [0.01, 0.30] | [0.01, 0.30] |

can stop at around the second iteration to ensure enough performance on force control, while PID need to be stop after the 9th iteration, and LADRC is around the 14th iteration). This can be explained by the fact that IMC has two parameters to tune, PID has three, and LADRC has four. The more parameters need tuning, the slower the convergence, and the more iterations required to reach the same low loss value. We tested the time required for a 10 s step force control simulation ten times, which was approximately 140.8 ± 5.6 ms (Mean and standard deviation). Considering that the DE algorithm sets the population size for each iteration to be at least four times the number of parameters $N_p$, the total optimization time $T_a = 4N_pI_a$. Therefore, the theoretical times are 3.38 s, 15.21 s, and 31.54 s for IMC, PID, and LADRC, respectively. The measured values were approximately 4.5 s, 16.2 s, and 32.3 s. The above results supports that leveraging a model-based design to simplify the controller may shorten parameter optimization time during pre-tuning of controller. Finally, practical tests were conducted on sheep's spine with three optimized controllers, and the results are depicted in Fig. 9. It can be observed that, based on the identified deformation-force model and designed loss, the three different types of controllers achieved similar performance in touch force control. However, compared to IMC, the PID controller may exhibit a smaller overshoot (as shown in Fig. 9(b)'s zoom-in area), while LADRC may be more sensitive to noise (highlighted in red oval frame as shown in Fig. 9(c)). Thus, fully utilizing the identified model to design controllers can lead to a slight improvement in control performance.

The robotic surgical system lacks tactile feedback mechanism, and doctors cannot judge the surgical operation status through touch like traditional surgery. In addition, if the direction or angle of the surgical tool needs to be adjusted during the surgical process, it may require remeasurement and calibration, increasing the complexity and time cost of the surgery. The internal model controller can adapt to intraoperative changes by adjusting model parameters in real-time. In other words, it continuously updates the internal model using feedback to maintain an accurate representation of the actual operating environment. Clearly, this requires enhanced computational speed and reasoning capabilities. We can expect that by integrating artificial intelligence method [28], the controller is expected to gain stronger self-learning and predictive capabilities, enabling faster responses to intraoperative changes and improving surgical precision and safety.

## 4. Conclusion

Many surgeries require precise control of the operational force applied to soft tissues, a task often accomplished by clinical surgeons through clinical touch and experience. For example, precise control of the contact force on the spinal cord is crucial to prevent dura mater tearing during the dissection of ossified ligaments. However, the complex viscoelastic characteristics of tissues at the lesion site pose a significant challenge for accurately applying touch force. Inspired by the operating patterns of clinical surgeons, this paper introduces a safe touch force servo framework designed for the precise control of contact force on the spinal cord with a robotic system. The framework incorporates tissue model identification and preoperative optimization of the force controller. Through theoretical analysis, simulation testing, and experimental validation, the effectiveness of this approach in achieving precise force control on the spinal cord has been verified. The proposed framework is extendable to apply precise forces to various vulnerable soft tissues, demonstrating its potential for widespread applications.

In the future, improvements to the soft tissue model will be made to consider more complex physiological characteristics, such as non-linearity and temperature sensitivity. Additionally, the proposed touch force servo framework will be optimized through exploring broader clinical experiments.

## CRediT authorship contribution statement

**Guangming Xia:** Writing – original draft, Data curation, Conceptualization. **Zifeng Jiang:** Validation, Data curation. **Bin Yao:** Data curation. **Yu Dai:** Writing – original draft, Project administration, Funding acquisition.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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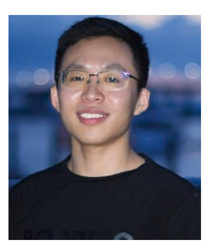
**Guangming Xia** received his B.Sc. degree from Harbin Engineering University, Harbin, China in 2019.

He studied as a visiting student at the National University of Singapore for one year from December, 2022. He is currently working toward the Ph.D. degree in control science and engineering with Nankai University, Tianjin, China.

His research interests include medical robots, trajectory planning of orthopedic robot, intelligent perception, precise motion control, and complex signal processing.

**Zifeng Jiang** received his B.Sc. degree from the College of Artificial Intelligence, Nankai University, Tianjin, China in 2022.

He is currently pursuing his Master degree at the Viterbi School of Engineering, University of Southern California, Los Angeles, California, United States. His research interests include medical robots, signal processing, image recognition and machine learning.



**Bin Yao** received the B.S. degree in automation from the Shandong University of Science and Technology, Qingdao, China, in 2018.

He studied as a visiting student at the National University of Singapore and Singapore General Hospital for one year from September, 2022. He is currently working toward the Ph.D. degree in control science and engineering with Nankai University, Tianjin, China.

His research interests include medical robots, force sensing and control, fiber optic sensing, signal conditioning circuit design, and signal processing.

**Yu Dai** received the M.S. and Ph.D. degrees in instrument science and technology from Harbin Institute of Technology, Harbin, China, in 2004 and 2009, respectively.

He is currently a Professor with the College of Artificial Intelligence, Nankai University, Tianjin, China. His research interests include medical robots and signal processing.